CHANTAL PELLEGRINI

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EDUCATION

PhD Candidate in Computer Science - Technical University of Munich *Nov 2022 - tentatively Nov 2026*
Focus: Multimodal foundation models and LLMs for medical AI
Advisor: Prof. Nassir Navab
M.Sc. in Computer Science - Technical University of Munich, Grade: 1.2 (top 10 %) *Apr 2020 - Jul 2022*
Exchange Semester - National University of Singapore *Aug 2018 - Dec 2018*
B.Sc. in Computer Science - Technical University of Munich, Grade: 1.5 (top 10 %) *Oct 2016 - Mar 2020*
studium MINT (Certificate in STEM subjects) - Technical University of Munich *Apr 2016 - Sep 2016*
German Abitur - Max-Planck High School Munich *2007 - 2015*

EXPERIENCE

Apple June 2025 - Sep 2025
AI/ML Intern *New York City, USA*

- Developed a LVML-based error detection system for Siri Visual responses.
- Designed zero-shot foundation model systems as well as fine-tuning-based approaches for automated quality assessment.
- Deployed final tool as web server and integrated in CI/CD pipeline for automated checks.

Technical University of Munich Nov 2022 - tentatively Oct 2026
PhD Candidate *Munich, Germany*

- **Multimodal foundation models**: Worked on large vision-language models combining visual information, structured representations, and language models for interactive conversation and reporting in radiology [7, 8] and operating room (OR) understanding. Our work on knowledge-enhanced foundation models in the OR received the MICCAI 2024 Best Paper Runner-Up Award [6].
- **Reinforcement Learning for LLMs**: Proposed a reinforcement learning method for calibrated confidence expression in large language models in factual question answering [1]. My ongoing work explores reinforcement learning for sequential decision making, learning in LLM-simulated clinical environments.
- **Longitudinal sequence modeling with LLMs**: Developed models that transform heterogeneous, irregular, multimodal event streams into structured language-like representations for forecasting and simulation over long temporal horizons [2].
- **Image Understanding**: Developed methods for fine-grained and interpretable medical image understanding, with a focus on explainable zero-shot X-ray diagnosis and structured radiology reporting via hierarchical visual question answering [4, 5].
- **Teaching and supervision**: Teaching courses in deep learning for medical imaging and supervising bachelor's and master's theses as well as practical courses.

Itestra GmbH Feb 2020 - Aug 2020
Student Researcher *Munich, Germany*

- Analysis of the usefulness of source code comments using classical machine learning and deep learning-based approaches in natural language processing.
- Development of a Visual Studio Code plugin to show the analysed comment usefulness while coding

Qualicen GmbH Feb 2019 - Jan 2020
Software Developer *Munich, Germany*

- Backend development for automated requirements analysis.
- Development of an automated testing framework for the company's software.

AWARDS AND SERVICES

- Best Paper Runner-Up (2nd place) in MICCAI 2024
- Nominated for the Apple Scholars in AI/ML PhD Fellowship 2024
- Acceptance and participation at the ICVSS 2024 Summer School
- Presentation at the open house of the Bavarian Academy of Sciences and Humanities 2024
- Reviewer for top conferences such as MICCAI, IPMI and ECCV
- Travel Grant for MICCAI 2024 in Marrakesh
- Travel Grant for MICCAI 2023 in Vancouver
- best.in.tum award 2022 - promotional program offered by the Department of Informatics of the TU Munich to the top 2% of their students
- Winner of Bertelsmann Robotics Hackathon 2018
- German Government Scholarship 2017-2018

SKILLS

Machine Learning	Multimodal LLMs, supervised and reinforcement learning, uncertainty calibration, timeseries forecasting, computer vision
Programming Languages and Frameworks	Python, PyTorch, Agentic coding, Java, C++, MATLAB
Languages	German (native), English (fluent), Spanish (basic)

SELECTED PUBLICATIONS

[1] David Bani-Harouni*, **Chantal Pellegrini***, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Nassir Navab, and Matthias Keicher. Rewarding doubt: A reinforcement learning approach to calibrated confidence expression of large language models. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2025.

[2] **Chantal Pellegrini**, Ege Özsoy, David Bani-Harouni, Matthias Keicher, and Nassir Navab. From EHRs to patient pathways: Scalable modeling of longitudinal health trajectories with LLMs. *arXiv preprint arXiv:2506.04831*, 2025.

[3] David Bani-Harouni, **Chantal Pellegrini**, Ege Özsoy, Matthias Keicher, and Nassir Navab. Language agents for hypothesis-driven clinical decision making with reinforcement learning. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2025.

[4] **Chantal Pellegrini**, Matthias Keicher, Ege Özsoy, and Nassir Navab. Rad-restruct: A novel vqa benchmark and method for structured radiology reporting. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 409–419. Springer, 2023.

[5] **Chantal Pellegrini***, Matthias Keicher*, Ege Özsoy*, Petra Jiraskova, Rickmer Braren, and Nassir Navab. Xplainer: From X-ray observations to explainable zero-shot diagnosis. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 420–429. Springer, 2023.

[6] Ege Özsoy*, **Chantal Pellegrini***, Matthias Keicher, and Nassir Navab. Oracle: Large vision-language models for knowledge-guided holistic OR domain modeling. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 455–465. Springer, 2024.

[7] **Chantal Pellegrini**, Ege Özsoy, Benjamin Busam, Benedikt Wiestler, Nassir Navab, and Matthias Keicher. Radialog: Large vision-language models for X-ray reporting and dialog-driven assistance. In *Medical Imaging with Deep Learning (MIDL)*, 2025.

[8] **Chantal Pellegrini***, Ege Özsoy*, Florian T Gassert, Alexander W Marka, Maximilian Strenzke, Matthias Keicher, Marcus R Makowski, and Nassir Navab. Enhancing radiology workflows through collaborative ai-assisted chest x-ray reporting using large vision-language models: a proof-of-concept study. *Insights into Imaging*, 17(1):123, 2026.

[9] **Chantal Pellegrini**, Nassir Navab, and Anees Kazi. Unsupervised pre-training of graph transformers on patient population graphs. *Medical Image Analysis*, 89:102895, 2023.

[10] Ege Özsoy, **Chantal Pellegrini**, Tobias Czempiel, Felix Tristram, Kun Yuan, David Bani-Harouni, Ulrich Eck, Benjamin Busam, Matthias Keicher, and Nassir Navab. MM-OR: A large multimodal operating room dataset for semantic understanding of high-intensity surgical environments. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 19378–19389, 2025.

* *denotes shared first authorship*